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Fermion Currents in Two-Dimensional Models

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Abstract. Using the path-integral approach we study abelian and non-abelian two-dimensional models. By computing fermion currents we discuss the role of the chiral anomaly in phenomena like fractionization of the fermion number, appearance of Wess–Zumino terms in QCD_2, $SU(N)$-Thirring model, etc.

Two-dimensional field theories recently revealed to be of particular relevance in real life. On one hand, the curious behavior of quantum systems in the presence of solitons, known as fractionization of the fermion number [1–6] can be understood by studying two-dimensional fermionic models relevant in condensed matter physics.

But also two-dimensional gauge theories, like QCD_2, appart from its intrinsic interest, have potential importance in connection with GUT monopoles since, due to spherical symmetry, they can be treated as effective two-dimensional systems [7–9].

The relevance of chiral anomalies in these phenomena and its relation with (global) topological properties of the models is by now well established.

When non-abelian symmetries are present, however, the usual field theoretical treatment of these problems, based on two-dimensional bosonization, becomes very involved. It is only very recently that non-abelian bosonization in the operator language has begun to be understood [10, 11]. Surprisingly enough, a Wess–Zumino functional [12] appears in this context, thus establishing another contact with more realistic models, namely low-energy effective actions used to study QCD current algebra [13, 14].

There is a path-integral approach to bosonization which has shown to be particularly adequate when non-abelian symmetries are present [15]. This approach is based in K. Fujikawa remarkable result on the non-invariance of the fermionic path-integral measure under chiral transformations [16]. In the present note we use this approach to study the above mentioned aspects of two-dimensional models, namely the role of the anomaly in phenomena like fractionization, appearance of Wess–Zumino terms in QCD_2, $SU(N)$-Thirring model, etc. As we shall see, it is very simple in the path-integral frame-work to compute fermion currents and analyze their properties both in the abelian and non-abelian cases.

We start by studying the simple 2-dimensional fermion model considered in [4–6], with (Euclidean) Lagrangian

$$\mathscr{L} = \bar{\psi}(i\partial\!\!\!/ + g e^{-\gamma_5 \theta})\psi, \tag{1}$$

where θ is a pseudoscalar soliton field which provides the back-ground inducing unusual quantum numbers. The (Euclidean) Dirac matrices satisfy the following relations:

$$\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}, \quad \gamma_5 = i\gamma_0\gamma_1, \quad \gamma_\mu\gamma_5 = i\varepsilon_{\mu\nu}\gamma_\nu,$$
$$\varepsilon_{01} = -\varepsilon_{10} = 1$$

As we shall see, one can very easily analyze the quantum theory around these different vacua within the path-integral approach by performing a chiral transformation and computing, *à* la Fujikawa, the associated Jacobian.

We want to compute the conserved fermion number current $\bar{\psi}\gamma_\mu\psi$ for Lagrangian (1) in the presence of θ. We then define the generating functional:

$$Z[s] = \int \mathscr{D}\bar{\psi}\mathscr{D}\psi \exp\left(-\int \bar{\psi}(i\partial\!\!\!/ + s\!\!\!/ + g e^{-\gamma_5\theta})\psi\, d^2x\right), \tag{2}$$

where we have introduced a source term s_μ in order to calculate the fermion current:

$$J_\mu(x) = \langle \bar{\psi}(x)\gamma_\mu\psi(x)\rangle = -\frac{1}{Z}\frac{\delta Z}{\delta s_\mu(x)}\bigg|_{s_\mu=0}. \tag{3}$$

In order to eliminate the γ_5-exponential in (2) we

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perform a chiral rotation:

$$\psi = \exp\left(\gamma_5 \frac{\theta}{2} t\right)\chi(x)$$
$$\bar\psi = \bar\chi(x)\exp\left(\gamma_5 \frac{\theta}{2} t\right), \tag{4}$$

where t is a parameter varying from 0 to 1. The Lagrangian in (2) now reads:

$$\mathscr{L}_s = \bar\psi(i\not\partial + \not s + g e^{-\gamma_5\theta})\psi = \bar\chi \mathscr{D}_t \chi, \tag{5}$$

with

$$\mathscr{D}_t = (i\not\partial + \not s + \tfrac{1}{2} t \gamma_\mu \varepsilon_{\mu\nu}\partial_\nu\theta + g e^{\gamma_5\theta(t-1)}). \tag{6}$$

Note that for $t=1$ we get rid of the γ_5-exponential.

It is known after Fujikawa [16] that the transformation (4) has a non trivial associated Jacobian. This Jacobian depends on the actual operator appearing in the action. In the present case, as we shall see, there is an additional term compared with the result of a massless model. Indeed, one can show following for example [17] that the Jacobian associated with transformation (4), for $t=1$,

$$\mathscr{D}\bar\psi\mathscr{D}\psi = \mathscr{J}(\theta)\mathscr{D}\bar\chi\mathscr{D}\chi, \tag{7}$$

is given by the expression

$$\mathscr{J}(\theta) = \exp\left[-\int d^2x \int_0^1 dt\, \mathrm{tr}(W_t\theta)\right], \tag{8}$$

where

$$W_t = -\frac{1}{4\pi}\gamma_5 \mathscr{D}_t^2. \tag{9}$$

Now, for the operator $\mathscr{D}_t$ defined in (6) we get:

$$\log\mathscr{J} = \frac{1}{2\pi}\int d^2x\left[s_\mu\varepsilon_{\mu\nu}\partial_\nu\theta + \frac{1}{4}(\partial_\mu\theta)^2 - \frac{g^2}{2}(\cosh 2\theta - 1)\right] \tag{10}$$

We remark that had we (incorrectly) computed the Jacobian defined in (4) just ignoring the last term in D_t, the cosh-term (10) would be absent. Indeed, this last omission is responsible for this absence in [5] and [18].

In terms of the new variables the generating functional reads:

$$Z = \exp\left[\frac{1}{2\pi}\int\left(s_\mu\varepsilon_{\mu\nu}\partial_\nu\theta + \frac{1}{4}(\partial_\mu\theta)^2 - \frac{g^2}{2}(\cosh 2\theta - 1)\right)d^2x\right] \cdot \int \mathscr{D}\bar\chi\mathscr{D}\chi \exp\left[-\int\bar\chi(i\not\partial + \not s + \not a + g)\chi d^2\right]. \tag{11}$$

where $a_\mu = \frac{1}{2}\varepsilon_{\mu\nu}\partial_\nu\theta$. Then, the fermion number current is given by

$$J_\mu(x) = \frac{-1}{2\pi}\varepsilon_{\mu\nu}\partial_\nu\theta + \frac{\delta}{\delta a_\mu(x)}\log\det(i\not\partial + \not a + g). \tag{12}$$

As we shall see, the first term in (12) is responsible for fractionization. The second one is just the gauge invariant current $j_\mu(x)$ for the χ-fields. We shall then write:

$$J_\mu(x) = -\frac{1}{2\pi}\varepsilon_{\mu\nu}\partial_\nu\theta + j_\mu(x). \tag{13}$$

Due to the non-triviality of the Jacobian (and hence to the chiral anomaly) we see that J_μ has been shifted due to the presence of the θ-term. This is nothing but the manifestation of the Schwinger terms [19, 20] in the current commutators

$$[J_0^5(x), J_0(y)] = \frac{-i}{\pi}\delta^1(x-y), \tag{14}$$

as can be very easily seen by passing to the operator language [21].

It is worthwhile to note that (11) can be used to the identity:

$$\det(i\not\partial + g e^{-\gamma_5\theta}) = \det(i\not\partial + g) \cdot\exp\left[\frac{g^2}{4\pi}\int d^2x(1-\cosh 2\theta)\right] \tag{15}$$

first obtained in [22] in an analysis of massive QED_2. Indeed, for $s=0$, Z can be written as follows:

$$Z[s=0] = \det(i\not\partial + g e^{-\gamma_5\theta}) = \exp\frac{1}{2\pi}\int d^2x\left[\frac{1}{4}(\partial_\mu\theta)^2 - \frac{g^2}{2}(1-\cosh 2\theta)\right]\det(i\not\partial + \not a + g) \tag{16}$$

Now, the determinant in the r.h.s. of (16) coincides with the fermionic part of the generating functional for massive QED_2 when the "gauge field" a_μ is taken in the Lorentz gauge. As it is well known, bosonization of this model yields to a Sine–Gordon theory and this result can be summarized, in the Path-integral formulation, as the following identity between the fermion determinant and the exponential of a bosonized effective action:

$$\frac{\det(i\not\partial + \not a + g)}{\det(i\not\partial + g)} = \exp\left[\frac{-1}{2\pi}\int d^2x \cdot\left[\frac{1}{4}(\partial_\mu\theta)^2 - g^2(1-\cosh 2\theta)\right]\right] \tag{17}$$

One can derive identity (17) by following the techniques described in [17]. Of course, one can apply these techniques directly to the determinant in the l.h.s. of (15) and this is, in fact, what is described in

[22]. Note that we are working in Euclidean space. Continuation to Minkowski space implies $\theta \to i\theta$ and a cosine term is gotten in (17).

Inserting (17) in (16) one finally gets the relation (15), of interest in the solution of many other two-dimensional models like, for example, the Gross–Neveu one.

In order to discover fractionization we now consider slow-varying θ-field. It is then easy to show that [4–6]:

$$j_\mu = \varepsilon_{\mu\nu}\partial_\nu\left[\text{const.}\frac{1}{g^2}\Box\theta + \text{higher order terms in } \Box\theta\right].$$

To leading order in derivatives of θ we then get:

$$J_\mu(x) \simeq \frac{-1}{2\pi}\varepsilon_{\mu\nu}\partial_\nu\theta. \tag{18}$$

Then, for a soliton with θ varying from O to π the fermion number is $-1/2$:

$$\mathcal{Q} = \int_{-\infty}^{\infty} J_0\, dx_1 = \frac{-1}{2\pi}[\theta(\infty) - \theta(-\infty)] = \frac{-1}{2}. \tag{19}$$

Of course one can get any other value just by changing the conditions at $x_1 = \pm\infty$.

Let us now extend our treatment to the non-abelian case. We can consider a purely fermionic model (like the $SU(N)$-Thirring or the chiral Gross–Neveu models) or a gauge theory like QCD_2. In the former case one uses a gaussian identity to pass from a purely fermionic interaction:

$$\mathcal{L}_{\text{int}} = \frac{-g^2}{N}(\bar{\psi}\gamma_\mu t^a\psi)^2. \tag{20}$$

(t^a being the generators of $SU(N)$ or $U(N)$ for the Thirring and Gross–Neveu models respectively) to an effective theory with Lagrangian:

$$\mathcal{L} = \bar{\psi}(i\not{\partial} + \frac{g}{\sqrt{N}}\not{A})\psi + \tfrac{1}{2}A^a_\mu A^a_\mu, \tag{21}$$

where A^a_μ is an auxilliary vector field (see [23] for details).

For the QCD_2 case the Lagrangian reads:

$$\mathcal{L}_{QCD_2} = \bar{\psi}(i\not{\partial} + e\not{A})\psi + \tfrac{1}{2}\operatorname{tr}F^2_{\mu\nu}. \tag{22}$$

Two differences should be noted between Lagrangians (21) and (22). For the Thirring model case, there is no kinetic term for A_μ. Instead, there is a vector field with $2(N^2-1)$ degrees of freedom ($2N^2$ for the $U(N)$ case). As a result, in the bosonized theory the bosons are massless. On the contrary, for QCD_2, being A_μ a gauge field and the $F^2_{\mu\nu}$-term present, the N^2-1 bosons get a mass (the Schwinger mechanism takes place [15]).

We shall now sketch how one can straightforwardly extend the technique developed above for the non-abelian case. For simplicity we shall consider QCD_2 but the Lagrangian (22) or the non-abelian extension of (1) can be similarly treated.

As before, we add to the Lagrangian (22) a source term:

$$\mathcal{L}_{QCD_2} = \bar{\psi}(i\not{\partial} + e\not{A} + \not{s})\psi + \tfrac{1}{2}\operatorname{tr}F^2_{\mu\nu}, \tag{23}$$

and then define the generating functional, this time depending on a non-abelian source. We then perform a non-abelian chiral transformation

$$\begin{aligned}\psi &= \exp(\gamma_5\phi t)\chi \equiv \mathcal{U}_5(t)\chi\\ \bar{\psi} &= \bar{\chi}\mathcal{U}_5(t), \quad \phi = \phi^a t^a\end{aligned} \tag{24}$$

Again, for $t=1$ the fermions decouple from A_μ provided one works in the "decoupling gauge" [15, 24] where any A_μ can be written in the form:

$$\not{A} = \frac{-i}{e}[\not{\partial}\exp(\gamma_5\phi)]\exp(-\gamma_5\phi). \tag{25}$$

the extension to an arbitrary gauge is straightforward).

The associated Jacobian can be computed from (8) except that now tr means both γ-matrices and color trace. The resuls is:

$$\log\mathcal{J} = \frac{-1}{4\pi}\int d^2x\, \operatorname{tr}^c[\tilde{s}^2_\mu - (s_\mu + eA_\mu)^2] + W_2[\phi, s], \tag{26}$$

$$\tilde{\not{s}} = \mathcal{U}_5(1)\not{s}\,\mathcal{U}_5(1).$$

where W_2 is the analogue of the Wess–Zumino functional [12] in two dimensions. (Its explicit form will be given below).

The derivation of (26) follows the method described in [17]. For the QCD_2 case, the details are given in [15]; for other two-dimensional models, the same results can be found in [25–26]. We shall only sketch the principal steps leading to (26). First, consider for simplicity the case $s=0$. The generalization of (8) to the non-abelian case reads:

$$\log\mathcal{J} = \frac{-1}{2\pi}\int d^2x\int_0^1 dt\, \operatorname{Tr}\gamma_5\mathcal{D}_t^2\phi \tag{27}$$

where Tr denotes both γ-matrices and color trace. $\mathcal{D}_t$ is given by:

$$\mathcal{D}_t = e^{\gamma_5\phi t}(i\not{\partial} + e\not{A} + \not{s})e^{\gamma_5\phi t} \tag{28}$$

One can easily show (see, for example [15]) that the Jacobian becomes

$$\log\mathcal{J} = \frac{e^2}{2\pi}\int d^2x\int_0^1 dt\, \operatorname{Tr}\gamma_5\phi(\not{\partial}\not{A}_t - ie\not{A}_t\not{A}_t) \tag{29}$$

where

$$\not{A}_t = \frac{-i}{e}(\not{\partial}e^{\gamma_5(1-t)\phi})e^{-\gamma_5(1-t)\phi}. \tag{30}$$

If we define:

$$\mathcal{U}(\phi, t) = \exp(2t\phi), \tag{31}$$

one can then show that:

$$(A_t)_\mu = (\tilde{V}_t)_\mu - \varepsilon_{\mu\nu}(\tilde{A}_t)_\nu \tag{32}$$

where

$$(\tilde{V}_t)_\mu = \frac{-i}{2e}[\partial_\mu \mathscr{U}^{1/2}(\phi,t), \mathscr{U}^{1/2}(\phi,t)]_-,$$
$$(\tilde{A}_t)_\mu = \frac{1}{2e}[\partial_\mu \mathscr{U}^{1/2}(\phi,t), \mathscr{U}^{1/2}(\phi,t)]_+ \tag{33}$$

Note that there is no γ_5 in the decomposition (32). This is due to the fact that, in $d=2$ dimensions, $\gamma_\mu \gamma_5 = i\varepsilon_{\mu\nu}\gamma_\nu$. Were we working in $d=4$, $\tilde{V}_t$ would correspond to the vector part and A_t to the axial part of A_t. (This is of course valid for $t=1$, that is, for the vector field in the decoupling gauge, (23)).

Now, using the following relations between $\tilde{A}_t$ and $\tilde{V}_t$ (They can be easily established, see for example [25–26]):

$$\partial_\mu \tilde{V}_{t\nu} - \partial_\nu \tilde{V}_{t\mu} - ie[\tilde{V}_{t\mu}, \tilde{V}_{t\nu}] + ie[\tilde{A}_{t\mu}, \tilde{A}_{t\nu}] \tag{34}$$

$$\partial_\mu \phi - ie[\tilde{A}_{t\mu}, \phi] = \frac{e}{2}\frac{\partial \tilde{A}_{t\mu}}{\partial t} \tag{35}$$

after some algebra one gets for $\log \mathscr{J}$:

$$\log \mathscr{J} = \frac{1}{8\pi}\int d^2x\, \mathrm{tr}^c(\partial_\mu \mathscr{U}^{-1}(\phi,1)\partial_\mu \mathscr{U}(\phi,1))$$
$$-\frac{i}{4\pi}\int_0^1 dt \int d^2x\, \varepsilon_{\mu\nu}\, \mathrm{tr}^c\, \mathscr{U}^{-1}(\phi,t)\partial_t \mathscr{U}(\phi,t)\mathscr{U}^{-1}$$
$$\cdot(\phi,t)\partial_\mu \mathscr{U}(\phi,t)\mathscr{U}^{-1}(\phi,t)\partial_\nu \mathscr{U}(\phi,t). \tag{36}$$

where the second term is the analogue of the Wess–Zumino functional [12] in two dimensions.

For non-zero source, $s \neq 0$, the Jacobian takes a more complicated form since, instead of relation (30) one has:

$$A\!\!\!/_t = \frac{-i}{e}(\partial\!\!\!/ e^{\gamma_5(1-t)\phi}) e^{-\gamma_5(1-t)\phi} + e^{\gamma_5 \phi t} s\!\!\!/ e^{\gamma_5 \phi t} \tag{37}$$

One can still apply the technique described in [15] (In particular, one uses (2.32)–(2.34) of the first paper in that reference). One then gets (26) with $W_2[\phi, s]$ defined as:

$$W_2[\phi, s] = \frac{e^2}{4\pi}\int_0^1 dt \int d^2x\, \mathrm{Tr}\, \gamma_5 \phi A\!\!\!/_t A\!\!\!/_t \tag{38}$$

If one uses (24)–(26) to write Z in terms of the new fermion variables χ and then integrates over them, one gets for the fermionic part of the generating functional, Z_F,

$$\log Z_F = \frac{-1}{4\pi}\int d^2x\, \mathrm{tr}^c[\tilde{s}_\mu^2 - (s_\mu + eA_\mu)^2] + W_2$$
$$+ \log\det(i\partial\!\!\!/ + s\!\!\!/) \tag{39}$$

Of course, the complete generating functional for QCD_2 should include the $F_{\mu\nu}^2$ term. However, one can use the fermionic part, (39), if one wants to compute fermion currents in the presence of an external background field. For example, for slow varying fields, we get for the current:

$$J_\mu^a = \langle \bar{\psi}\gamma_\mu t^a \psi \rangle \tag{40}$$

the following expression

$$J_\mu^a = -\frac{e}{2\pi}A_\mu^a + \frac{\delta W_2}{\delta S_\mu^a}[\phi, s]\bigg|_{s=0} \tag{41}$$

We then see that the signal of the Schwinger mechanism ($J_\mu \sim A_\mu$) can be again discovered for the QCD_2 case: as explained by Swieca [2] for the abelian case, the presence of a term like the first one in (41) indicates that the gauge fields acquire a mass in a sort of intrinsic Higgs mechanism. The role of the anomaly in this phenomenon is clear in our approach: both terms in (41) are present due to the non-triviality of the Jacobian associated to the non-abelian chiral transformation (24). Concerning the Wess–Zumino contribution, it has been recently remarked [28–30] its relevance in the computation of fermion currents, in particular in 3-dimensional models. In particular, in [28–29] it is shown for particular field configurations (constant or static field strengths) that analogous terms appear, connected in this $d=3$ case with parity violation. Both our example and these $d=3$ ones show that preservation of gauge invariance (the current is covariantly conserved) induces violation of chiral ($d=2$) or parity ($d=3$) symmetries. Terms of topological origin then appear and may induce unexpected behavior of currents.

As an example, let us consider the simple $SU(2)$ case and take A_μ in the decoupling gauge. Then

$$\mathbf{A}_\mu = \frac{1}{e}\bigg[\varepsilon_{\mu\nu}(\partial_\nu|\phi|\check{\phi} + \mathrm{sh}|\phi|\partial_\nu\check{\phi})$$
$$-2\,\mathrm{sh}^2\frac{|\phi|}{2}(\check{\phi} \wedge \partial_\mu\check{\phi})\bigg], \tag{42}$$

where we have written:

$$\phi = |\phi|\frac{\boldsymbol{\sigma}}{2}\cdot\check{\phi}, \quad \check{\phi}\cdot\check{\phi} = 1 \tag{43}$$

Concerning the Wess–Zumino term, the derivative in (41) can be straightforwardly but laboriously computed giving a complicated expression in terms of ϕ. However, for the particular configuration $\check{\phi} = $ constant, this contribution vanishes and then we get for the fermionic current:

$$\mathbf{J}_\mu = \frac{-1}{2\pi}\varepsilon_{\mu\nu}\check{\phi}\partial_\nu|\phi| \tag{44}$$

We then see that a result similar to that of (18) is obtained. Indeed, being $\check{\phi} = $ constant we can define a charge $Q = \int \check{\phi}\cdot \mathbf{J}_0 dx_1$ and, for appropriate $|\phi|$ get a fractional answer. In this particular case our model has effectively reduced to the abelian case and the result (44) confirms the Schwinger mechanism.

In order to analyse the actual $SU(N)$ case and investigate if unusual effects are induced by non-trivial background fields, one should consider arbitrary fields and take into account the Wess–Zumino term. What our results have shown, and this will be our conclusion, is that the path-integral approach provides a very convenient framework in order to compute fermion currents, analyse fractionization, bosonization and, in general, all phenomena where preservation of an invariance (like gauge invariance in our last example) induces anomalies in those currents.

We expect that the techniques described above should prove to be very useful in studying other interesting problems like, for example, those arising in the Kondo Model. Work on this aspect is in progress.

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